
A TWO-QUARTER DECISION DOCUMENT

Spotify AI DJ

Two users. Same product. One wants to steer it.
One wants to be left alone.

On agency, foils, and the metric I would refuse to optimize for — a senior product decision document, built without access to internal Spotify data, on a feature shipped in February 2023 and still finding its shape.

AUTHOR

Aashi Thakkar

AI Product Manager

MMXXVI

AASHITHAKKAR.COM

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In six sections.

A case study reasons toward an answer. This one takes the loss-of-agency friction in Spotify's AI DJ and asks: *if I owned this surface for the next two quarters, what would I ship — and what would I refuse to optimize for in the process?*

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Evolve the existing thumbs into a richer signal layer, and surface them more prominently when the recommendation model's confidence is low — so the controls show up exactly when the algorithm is least sure.

SECTION ONE

Context

A feature shipped on a bet. A paper that named the cracks. And the friction this case study takes.

In February 2023, Spotify shipped AI DJ. A friendly voice. A curated stream. A bet that listeners would trade input for ease — that the algorithm would be good enough that telling it what you wanted would feel old-fashioned.

Nearly three years later, a paper out of Singapore Management University analyzed more than fourteen hundred user comments about the feature and found something quieter than backlash. Three frictions. Users felt they'd lost control. The voice broke its own promises — it said jazz, it played funk. And the music looped back on itself.¹

THIS STUDY

I.

LOSS OF AGENCY

Users feel the algorithm decides everything. They can like or dislike, but cannot steer.

II.

ANTHROPOMORPHISM GAP

The voice promises a mood and delivers a different one. Users build expectations of a relationship.

III.

REPETITION

Same songs, different sessions. The catalog feels smaller than it is.

This case study takes the first — *loss of agency* — and asks one question: if I owned AI DJ at Spotify, what would I ship in the next two quarters to fix it, without breaking

what passive listeners love about the feature today?

A NOTE ON FRAMING

I don't work at Spotify. This case study is a strategic exercise — a thought experiment in how I'd structure the decision, the metrics, and the risks if the problem landed on my desk. It is built without internal Spotify data. Section VI is honest about the questions I would need answered to commit to this roadmap with confidence. The point is not to be right about the answer. The point is to show the shape of how I would reason toward one.

SECTION TWO

Problem Framing

A problem is a specific user, with a specific job, hitting a specific wall — and someone who depends on the current behaviour *not* being broken to fix it. So this section talks about two people.

THE PROTAGONIST

Meet Maya.

26 · MARKETING COORDINATOR · AUSTIN

Spotify Premium for four years — the kind of user Spotify counts on: habitual, loyal, lightly engaged. When AI DJ launched, she tried it three times in week one. By week four, she had stopped opening it.

Maya is the user a real listener on Spotify's Ideas Exchange described herself as:

"It's so frustrating knowing the vibe I want, but Spotify's attempt at guessing what I want is always wrong. Every. Single. Time. I just want to TELL you what I want."

SPOTIFY COMMUNITY · IDEAS EXCHANGE · 2024

Maya's version of that — in her own words:

”

It's like a friend who only listens when I'm angry. I can downvote a song, but I can't tell it I want lo-fi today, not throwback hip-hop.

MAYA · ENGAGED-BUT-STUCK

Composite quote, synthesized from comment patterns in Mukherjee et al. (2025) and Spotify Community threads. Maya is a representative user, not a single source.

JOB TO BE DONE

When I open AI DJ in a specific mood, I want to nudge it without abandoning the surprise — so it stays mine without becoming work.

THE FOIL

Meet Jordan.

34 · DESIGN LEAD · THE MORNING COMMUTER

He listens to AI DJ on his morning commute and nowhere else. He doesn't skip. He doesn't re-prompt. He doesn't have an opinion. An XDA reviewer described his own version of this habit as "*the background noise of my day*"² — and that is exactly the bar AI DJ clears for Jordan. He is the user Spotify built the feature for. If we put visible controls on the player and Jordan suddenly has to think about whether to press them, the feature breaks for him before it fixes itself for Maya.

DESIGN CONSTRAINT · LOCKED

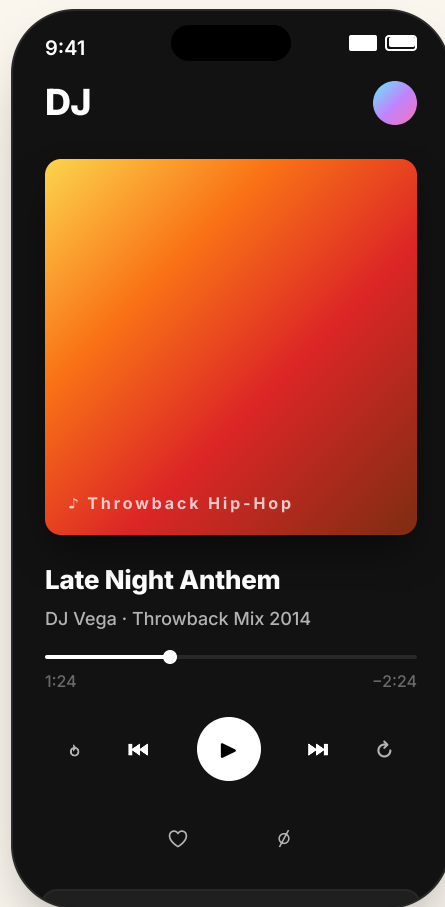
Option II has to work for Maya without breaking Jordan.

WHY NOW

AI DJ launched in early 2023. In late 2024, Spotify shipped text prompts — partial agency, but front-loaded: tell DJ what you want *before* the session, not during. The mid-session gap is still wide open, and it is where Maya lives every time she is three songs in and the mood is wrong. Competitors are circling. None have shipped this exact capability. The window to own it is probably two quarters.

COST OF INACTION

If we don't ship this, Maya does not come back in week five — she returns to her own playlists, and AI DJ quietly becomes a feature Spotify markets but Maya doesn't use. The cost isn't a churned subscriber. It's a paid subscriber who stopped trusting the most expensive AI surface on the platform.



i. **Binary signal only.**
Thumbs let users dislike, but can't say why or how much.

ii. **DJ talks; doesn't listen.**
The voice promises a mood but offers no mid-session input.

iii. **Wrong mood, no nudge.** Maya's only option is to skip the entire session.

Maya's agency in the current product ends at like, dislike, or skip. There is no surface for her to communicate what kind of mood she is in once a session is underway.

SECTION THREE

Solution Space

Three ways to give Maya agency without taking it from Jordan. Each is a real product. Each has been thought about by someone at Spotify already. Each wins or loses on a different cut.

I.

MOOD PROMPT AT SESSION START

CUTS

A text input that appears when AI DJ launches. Maya types "*lo-fi morning*" and DJ shapes the session around the prompt. Essentially what Spotify shipped in late 2024 — but front-loaded. Jordan gets asked his mood at 7:42 AM. Maya's friction is mid-session, not pre-session.

II.

EVOLUTION OF THUMBS & CONFIDENCE-AWARE SURFACING — *the pick*

SHIP

Existing thumbs evolve into a richer signal: long-press surfaces optional, non-blocking reason capture. Semantic shift from binary like/dislike to graduated "*more like this / less like this.*" Controls surface more prominently when the recommendation model's confidence is low.

III.

CONTEXT-AWARE AI DJ

LONG-TERM

DJ infers mood passively from signals it already has — time of day, recent listening, calendar context. Maya never has to ask. Jordan never has to notice. The right answer in a four-quarter window. Requires ML infrastructure Spotify does not have wired into AI DJ specifically.

WHY OPTION II WINS ON THE CUT

It is the only option that ships in the window without breaking Jordan. Option I ships, but breaks the passive default. Option III protects the passive default, but does not ship in time. Option II does both.

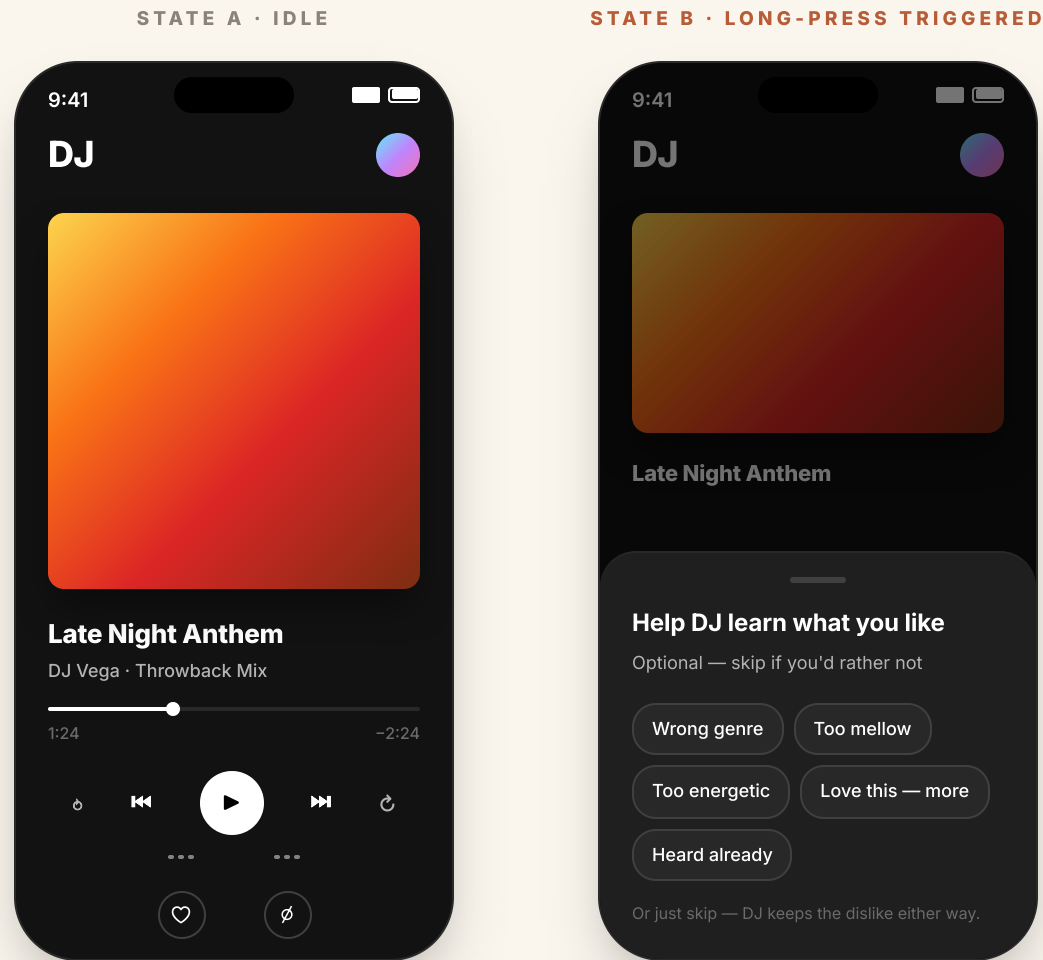
It builds on a surface users already understand. Thumbs on AI DJ tracks are two years old. The mental model exists. Adding semantic richness and dynamic prominence does not require teaching a new gesture — it deepens one that is already learned.

It generates the training signal Spotify needs for Option III. Every "*more like this*" with an optional reason becomes labeled training data for the contextual model that

powers Option III in 2027. We do not ship Option III this year. We ship the data infrastructure that makes Option III cheap to ship next year.

PHASING AND THE GATE

Phase 1 (first quarter). Evolve thumbs. Long-press surfaces optional reason capture. Ship to alpha at 5% of AI DJ DAU. **Phase 2 (second quarter).** Confidence-aware surfacing — controls visibility scales with inverse model confidence. Ship to beta at 25%, gated on Phase 1 outcomes. The phasing is real. Phase 2 is conditional, not automatic.



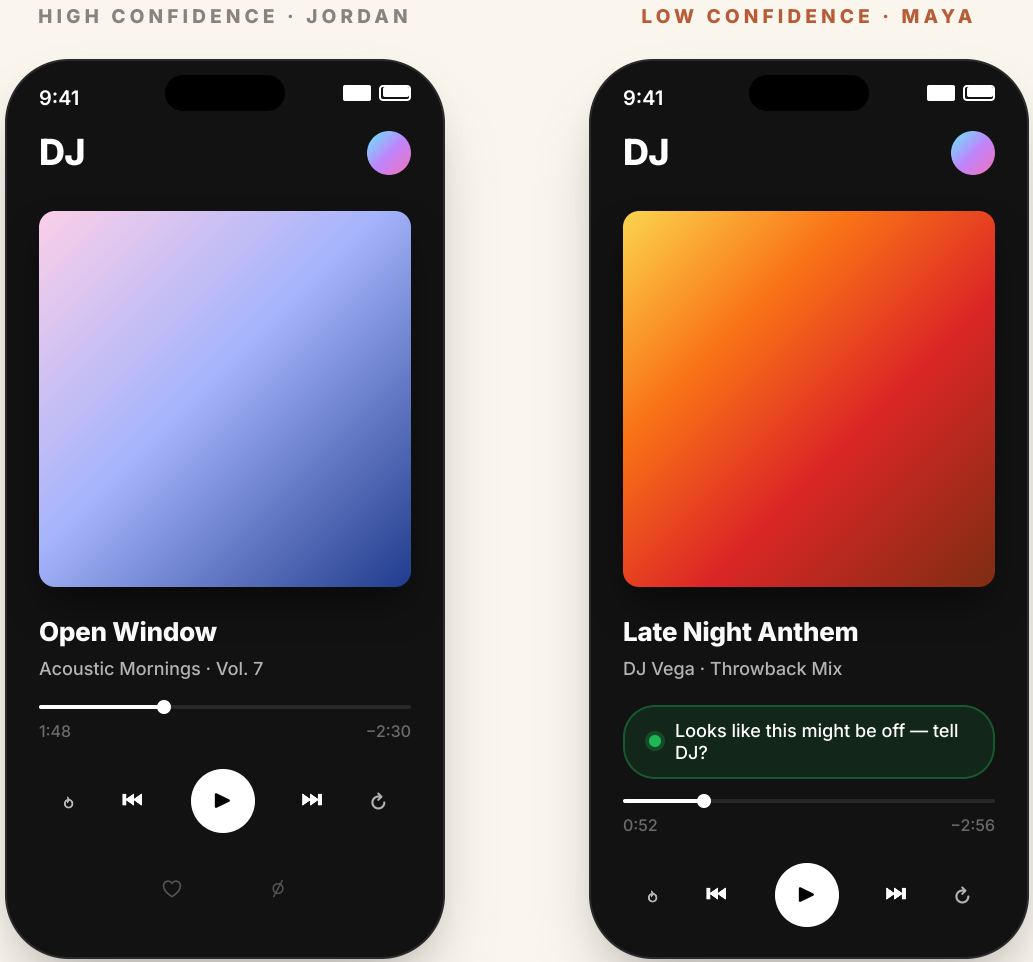
Subtle affordance.

Long-press extends existing thumbs — no new gesture to learn.

Half-sheet, never modal.

Reason capture is optional and dismissable.

The dislike is logged whether or not the user engages reason capture. The "Heard already" chip doubles as the Risk I detection mechanism.



Player stays calm.

Model is confident — controls visually receded.

Controls surface.

Ambient hint appears, thumbs become prominent.

The controls show up when the algorithm is least sure. Same gesture as Phase 1 — only visibility changes.

SECTION FOUR

Success Metrics

The metric question is uniquely tricky here. The real outcome — whether Maya comes back in week five — is far downstream from the action — whether she taps a control mid-session. So we need a proxy. And the wrong proxy will give us a feature that *looks* like it's working when it isn't.

WHAT I AM ACTUALLY TRYING TO MOVE

AI DJ week-4 DAU. The real outcome is retention. Specifically: does the user who tried AI DJ in week one still use it in week four? Maya's behaviour was the diagnosis — three sessions in week one, zero by week four. The feature succeeds if it changes that arc for users who look like her. Why I cannot ship against it: it is a four-week lagging indicator. We need a proxy that is faster — and honest.

THE PROXY I WOULD SHIP AGAINST

Mid-session control engagement. The percentage of AI DJ sessions in which the user interacts with any thumbs control after the first track. **Threshold for the Phase 1 → Phase 2 gate:** +50% relative lift versus the existing-thumbs baseline, with no degradation in guardrails. A +20% lift would read as "*users tolerated it.*" A +50% lift is the signal that the redesigned controls actually changed how users behave mid-session.



Each gate is a kill decision in disguise. A feature that passes Alpha but stalls at Beta gets pulled, not iterated indefinitely.

VOCABULARY · GOODHART'S LAW

"When a measure becomes a target, it ceases to be a good measure." A less-careful PM would ship against likes per session. It would also be wrong: when likes become the target, users start liking *defensively* — tapping thumbs-up to vote for songs they want more of, decoupling the signal from preference. The like stops meaning "this is good" and starts meaning "please remember this for next time."

THE FRAMEWORK, AT A GLANCE

SHIP AGAINST Mid-session control engagement +50% relative lift vs. control. Percentage of sessions where users interact with any thumbs control after the first track.	REFUSE TO OPTIMISE FOR <i>Likes per session</i> Easy to move, easy to game. The signal stops meaning preference and starts meaning "remember this."	GUARDRAILS Three kill conditions • Time-to-music ≤ +100ms (p95) • 60-second skip ≤ +1 pp • AI DJ NPS ≤ -2 points
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SECTION FIVE

Risks & What Could Go Wrong

Every case study has a section like this. Most of them are box-checking. This one names the four ways I might be wrong, and what we would do to catch each before it costs us.

RISK I

The diagnosis is wrong.

Maya's friction might not be agency — it might be repetition. Option II ships and does not move week-4 DAU.

DETECTION

Phase 1's reason-capture categories. If users overwhelmingly select *"heard this already,"* the framing was wrong.

MITIGATION

Pivot Phase 2 to confidence-aware *re-ranking*. Same engineering substrate, different target.

RISK III

Reason capture reads as surveillance.

"Why does Spotify need to know *why* I downvoted this?" If long-press feels intrusive, engagement collapses.

DETECTION

Reason-capture engagement among thumb users. If >85% tap thumbs but skip the reason layer, the framing is the problem.

MITIGATION

Copy, not mechanic. *"Help DJ learn what you like"* lands. *"Tell us why"* doesn't.

RISK II · LOAD-BEARING

Jordan churns silently.

If the new controls make Jordan feel like the player got noisier, he stops opening AI DJ. He doesn't complain. He drifts.

DETECTION

Segmented week-4 DAU. Track retention for users who did *not* engage with new controls during alpha.

MITIGATION

Phase 2 confidence-aware surfacing is the structural fix. If degradation shows before Phase 2 ships, pull and rework.

RISK IV

Confidence-aware UI feels arbitrary.

If controls appear and disappear unpredictably, the surface reads as buggy, not intelligent. Confidence-aware becomes confidence-*confusing*.

DETECTION

Variance in control visibility per session. If UI flips more than three times in 30 minutes, we are flickering.

MITIGATION

Smoothing thresholds — UI state changes every N tracks. Ambient never-modal explainability affordance.

SECTION SIX

Open Questions

Senior product work is not knowing every answer. It is knowing which unknowns change the call — and being honest about them in writing.

I.

What is the Jordan-to-Maya ratio in AI DJ DAU?

The case study is built on the premise that both users exist in meaningful numbers. If Jordan-types are 80% of DAU and Maya-types are 15%, the design has to bias much harder toward the invisible default than Phase 1 currently does.

WHAT I WOULD DO DIFFERENTLY

If Jordan is dominant: Phase 1 gets pared down — the long-press reason layer ships, but the prominent "more like this" player surface defers to Phase 2.

II.

Does Spotify expose model confidence at the recommendation layer?

Phase 2 assumes the recommendation team can expose a per-track confidence score. Modern recommenders usually do — but Spotify's AI DJ architecture might bundle generation and ranking in ways that make confidence implicit, not addressable.

WHAT I WOULD DO DIFFERENTLY

If confidence access is more than one quarter of work: Phase 2 becomes a separate Q3–Q4 bet. Phase 1 stands as a complete shipment.

III.

What is the current usage pattern of existing AI DJ thumbs?

"Evolution of thumbs" assumes thumbs are an established surface. If internal telemetry shows thumbs are essentially dead (<2% of sessions), we are putting new paint on a door no one walks through.

WHAT I WOULD DO DIFFERENTLY

If thumbs are dead: gesture-based controls — swipe up/down on player — a new surface unburdened by an unused one.

IV.

What did Spotify learn from the late-2024 text-prompt launch?

Option I was dismissed from public signal — thin promotion suggests retention did not move. But Spotify has the internal telemetry.

WHAT I WOULD DO DIFFERENTLY

If text-prompts *did* move retention internally: Option I isn't dead — it's complementary. Combine mid-session controls with a post-session prompt surface.

THE HONEST READ

Every question above is one a Spotify PM could answer in a thirty-minute conversation with the right person on the AI DJ team. The fact that this case study is built without those answers is exactly why it lives on a portfolio page, not in a roadmap doc — and naming that limitation in writing is the move.

APPENDIX

Sources

This case study is built without internal Spotify data. Every claim about user behaviour is grounded in publicly available research and primary user testimony.

¹ PRIMARY RESEARCH

Mukherjee, A., Chang, J., & Wibowo, S. (2025). Managing the personalization paradox: Lessons from Spotify's AI DJ.

Singapore Management University. Peer-reviewed publication, November 2025. Analysis of 1,400+ AI DJ user comments. Source for the three frictions framework.

USER TESTIMONY

Spotify Community · Ideas Exchange — "Make Spotify DJ less repetitive"

Public Spotify Community thread, 2024. Source for Maya's anchor pattern — direct user request for mood-based input and the explicit voice "I just want to TELL you what I want."

² USER TESTIMONY

XDA Developers — "I want to quit Spotify, but its AI DJ keeps pulling me back."

Published January 2026. Source for Jordan's pattern — the description of AI DJ as "the background noise of my day."

METHODOLOGY

Goodhart's law & the proxy-target trap in ML systems.

Foundational principle (Goodhart, 1975). See also: Manheim & Garrabrant (2018) on Goodhart variants in machine learning systems; Spotify engineering writing on recommendation A/B testing methodology.

PRODUCT TIMELINE

Spotify AI DJ — launches and product updates.

Initial launch: February 2023 (US & Canada Premium beta). Text prompt feature: late 2024. DJ Liv (Spanish-language): 2024.

COLOPHON

Aashi Thakkar

AI Product Manager building at the intersection of generative AI infrastructure and consumer experience. Previously a software engineer; now shipping multi-agent GenAI products to capital markets professionals at global scale. Writing the *AI for PMs* series on LinkedIn — explainers, teardowns, and decision documents at the AI-meets-consumer-PM boundary.

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WEB aashithakkar.com

LINKEDIN in/aashithakkar29

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